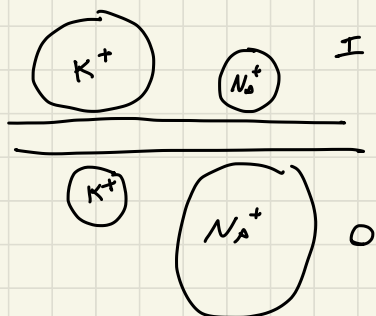


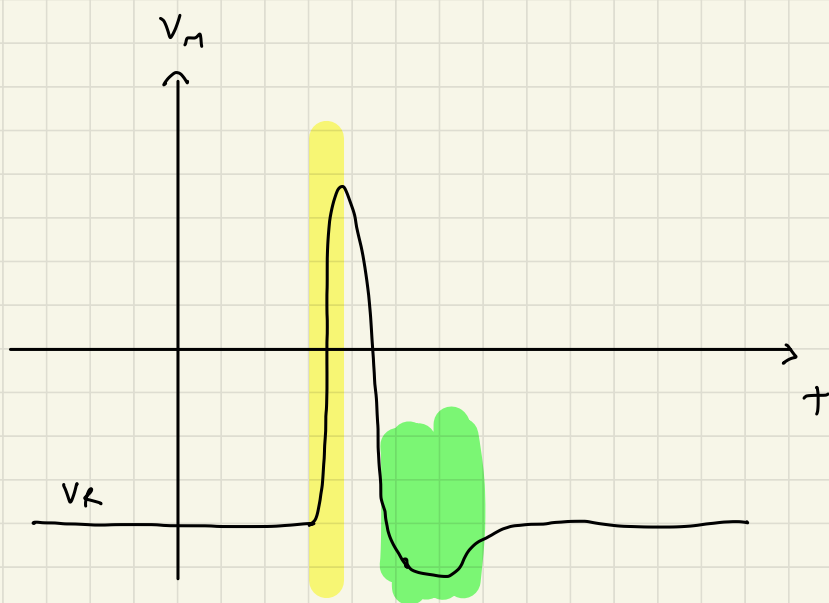
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$$G_{Na}(V_m, t)$$

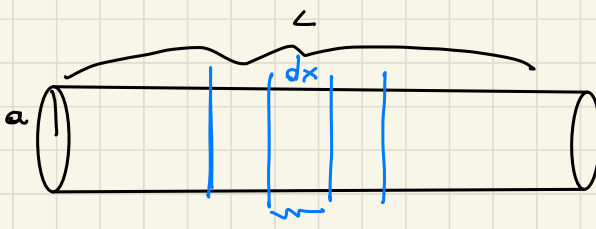
$$G_K(V_m, t)$$



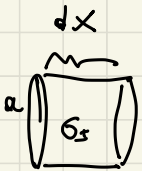
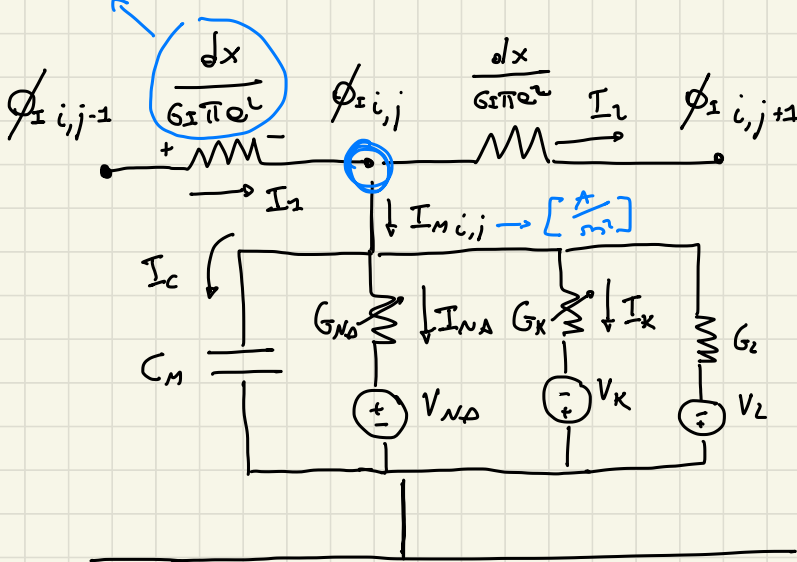
$$\left. \begin{array}{cc} [Na]_I & [Na]_O \\ [K]_I & [K]_O \end{array} \right\} \text{constant}$$



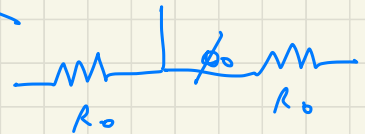
$\angle \gg \phi$



$[Z] \rightarrow I_1 [A]$ circuit... $i \rightarrow \text{time} \quad j \rightarrow \text{SPACE}$



$$\frac{1}{G_1} \frac{dx}{\pi Q^2}$$



$$R_0 \ll \frac{dx}{61 \pi Q^2}$$

$$\Rightarrow \phi_0 = C_0 \omega T$$

$$I_1 = \frac{\phi_{i,j-1} - \phi_{i,j}}{dx} G_{i,j} 2\pi Q^2$$

$$I_2 = \frac{\phi_{i,j} - \phi_{i,j+1}}{dx} G_{i,j} 2\pi Q^2$$

$$I_1 = I_2 + I_{M,i,j} 2\pi Q dx$$

$$\left[\frac{\phi_{i,j-1} - \phi_{i,j}}{dx} - \frac{\phi_{i,j} - \phi_{i,j+1}}{dx} \right] G_{i,j} 2\pi Q^2 = I_{M,i,j} 2\pi Q dx$$

$$I_{M,i,j} \left(\frac{2}{G_{i,j}} \right) = \frac{\phi_{i,j-1} + \phi_{i,j+1} - 2\phi_{i,j}}{dx^2} \left[I_M \frac{2}{G_{i,j}} = \frac{\tilde{\nabla} V_M}{dx^2} \right]$$

$$dx \rightarrow \phi$$

$$\frac{\partial^2 \phi(x,t)}{\partial x^2} = I_M(x,t) \frac{2}{G_{i,j}}$$

GENERAL!

$$I_M = I_C + I_{ionic} = C_M \frac{dV_M}{dt} + I_{ionic}$$

$$\frac{\partial^2 \phi_I}{\partial x^2} = \frac{2}{G_{I0}} \left(C_M \frac{\partial V_M}{\partial t} + I_{ionic} \right)$$

$$\frac{\partial V_M}{\partial t} = \frac{1}{C_M} \left(\frac{G_{I0}}{2} \frac{\partial^2 \phi_I}{\partial x^2} - I_{ionic} \right) \quad \text{GENERAL}$$

Hyp $\rightarrow \phi_0 = \text{const}$

$$V_M = \phi_I - \phi_0$$

$$\frac{\partial^2 V_M}{\partial x^2} = \frac{\partial^2 \phi_I}{\partial x^2}$$

H.H

$$\frac{\partial V_M(x,t)}{\partial t} = \frac{1}{C_M} \left(\frac{G_{I0}}{2} \frac{\partial^2 V_M(x,t)}{\partial x^2} - I_{ion}(x,t) \right)$$

$\rightarrow \phi_K \rightarrow \phi_{OA}$

$\rightarrow \text{NO} \rightarrow \text{STIMULATION}$

WITH EXT. CURRENT

NERVE CUFF
STIM.

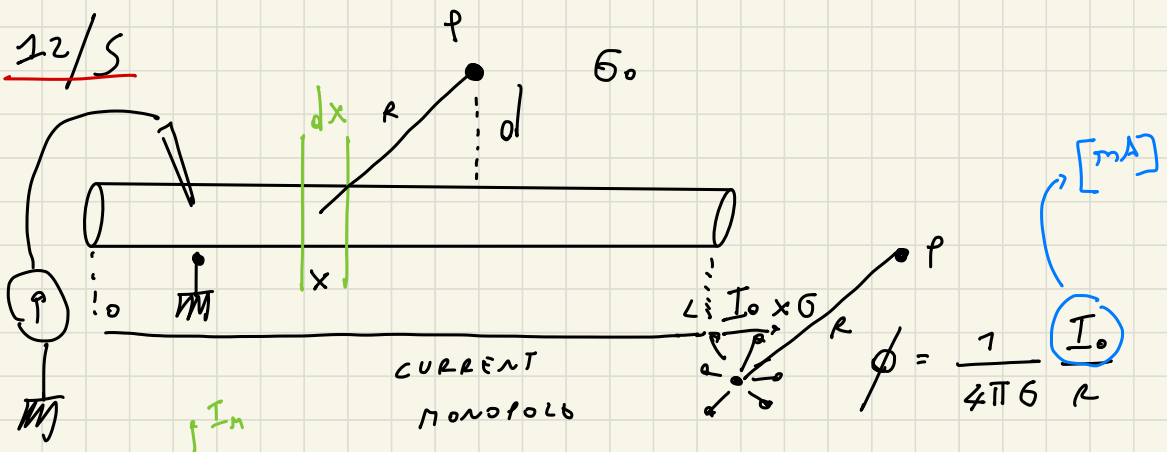
$\rightarrow \text{FEJ}$

INITIAL
CONDITIONS
GATING
VARIABLES

$\rightarrow$

$$\tau_0 = \frac{\alpha_N(V_R)}{\alpha_N(V_R) + \beta_N(V_R)}$$

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$\rightarrow d\phi = \frac{I_m(x,t) 2\pi a dx}{2 \epsilon_0 R}$

$$d\phi = \frac{I_m a dx}{2 \epsilon_0 R(x)}$$

$$\phi(t) = \frac{a}{2 \epsilon_0} \int_0^L \frac{I_m(x,t)}{R(x)} dx$$

$V_n \rightarrow I_n ?$

$$I_n = \frac{\sigma_1 a}{2} \frac{\partial \tilde{V}_n}{\partial x^2}$$

$$\phi(i) = \frac{a}{2 \epsilon_0} \sum_{j=1}^{M+1} \frac{I_{n,i,j}}{R_j} dx$$

$$\frac{\partial^2 \phi_I(x,t)}{\partial x^2} = I_n(x,t) \frac{z}{G_I Q}$$

$$I_n = C_n \frac{dV_n}{dt} + I_{ion}$$

$$\frac{\partial^2 \phi_I}{\partial x^2} = \frac{z}{G_I Q} \left(C_n \frac{dV_n}{dt} + I_{ion} \right)$$

$$\frac{dV_n}{dt} = \frac{1}{C_n} \left[\frac{G_I Q}{z} \frac{\partial^2 \phi_I}{\partial x^2} - I_{ion} \right]$$

$$V_n = \phi_I - \phi_o \quad \phi_I = V_n + \phi_o$$

$$\frac{dV_n(x,t)}{dt} = \frac{1}{C_n} \left[\frac{G_I Q}{z} \frac{\partial^2 V_n(x,t)}{\partial x^2} + \frac{G_I Q}{z} \frac{\partial^2 \phi_o(x,t)}{\partial x^2} \cdot I_{ion}(x,t) \right]$$

